Mathematical Stories

Lecture Notes

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Chapter 1

Introduction

1.1 Purpose and Philosophy

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognises the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, formulate precise questions, distinguish an object from its notation, notice structure, and build simple but meaningful chains of reasoning. A solved problem is valuable not only because it produces an answer, but also because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

The course is organised into three connected parts: linear algebra, analytic geometry, and calculus. In the first part, we focus on matrix methods: matrices, determinants, inverse matrices, and systems of linear equations. The second part develops the language of coordinates and vectors and uses it to describe points, lines, planes, directions, distances, and angles. The third part introduces sequences and functions, limits, continuity, derivatives, integrals, change, and accumulation. These parts are not isolated collections of techniques. Together they form a basic mathematical language for describing structure, space, and change.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorise, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as an answer to a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some habits of thought will deliberately return in different places. The same question of what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. A mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognise patterns of reasoning rather than only patterns of exercises.

The number of answers produced is not, by itself, a measure of progress. What matters is the quality of the work. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each important step was valid? Did we interpret or verify the result? These questions are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics as the art of asking good questions and giving well-reasoned answers.

1.2 How the Course Works

This book is part of a weekly learning process rather than a collection of material to be read only before an examination. Each of the twelve thematic chapters is designed to support one week of study. A chapter provides the mathematical structure for the lecture, a complete narrative to which the student can return, and a starting point for individual problem solving and classroom discussion.

In a fifteen-week semester, the twelve thematic weeks establish the main rhythm of the course. The remaining time can be used to introduce the method of work, connect topics, revise important ideas, and present more substantial solutions.

1.2.1 The Weekly Learning Cycle

A student should not leave a topic behind when the lecture ends. A typical week consists of the following connected stages:

1. **Attend the lecture.** The lecture introduces the topic, motivates the central questions, and demonstrates how the relevant mathematical objects and methods should be read.

2. **Return to the chapter.** Read the corresponding chapter carefully after the lecture. The PDF is not merely a summary of slides. It is an organised narrative containing definitions, explanations, examples, interpretations, and mathematical arguments that can be examined more than once.

3. **Ask questions and clarify the material.** Mark every place where a definition, transition, calculation, or conclusion is not yet clear. Use the text, your own examples, classroom discussion, and appropriate digital tools to resolve these questions.

4. **Solve the complete assigned problem set.** Move from following an explanation to constructing arguments independently. Work on every assigned problem and prepare a separate, self-contained note for each one.

5. **Prepare professional mathematical notes.** Use precise notation, explanatory sentences, justified transitions, and a clear conclusion. The note should show not only the answer, but also how it follows from the given information, definitions, and earlier results.

6. **Present and discuss your work.** During exercise classes, be prepared to explain a solution, answer questions about its steps, and describe what you learned while preparing it.

These stages form one continuous process. Each stage prepares the next, and the presentation and feedback lead back to further reading, revision, and improved work.

1.2.2 Exercise Classes: Presentation, Feedback, and Revision

Presenting mathematics is part of learning mathematics. It reveals whether an argument has genuinely been understood and whether the student can communicate it clearly. During a presentation, the student should be able to read the notation, explain the role of each important formula, justify the method, and respond to questions without relying on AI to speak on the student's behalf.

Feedback given during exercise classes is intended for the whole group, not only for the student whose work is currently being presented. Everyone should follow the argument and listen carefully to the discussion. By comparing different examples, students learn what a complete solution looks like, which explanations make an argument clear, which steps require justification, and which weaknesses require revision.

Students should not focus only on comments addressed directly to their own work. Listening to feedback on the work of others provides a broader set of examples and develops an operational understanding of the course standards: what makes mathematical work clear, justified, and complete, and what still needs improvement.

The feedback concerns the mathematical work and its results, not the personal worth of the student. What is examined is the quality of the note, the understanding demonstrated in the argument, and the clarity of the presentation. Feedback should identify what already works and what should be changed in order to produce a stronger solution.

The first submitted version does not always have to be the final version. Students should learn to receive comments, reconsider their choices, correct errors, add missing explanations, and improve both the written note and its presentation. Revision is not evidence of failure. It is a normal part of serious intellectual work.

This process is important throughout university study and later in professional life. Students and professionals are given tasks, asked to solve problems, expected to present results, and required to respond to feedback and critical review. Learning to improve an initial solution is therefore not an additional classroom exercise. It is a fundamental habit of responsible academic and professional work.

1.2.3 Progress and Assessment

Progress is demonstrated by the quality and increasing independence of the student's work. By the end of a weekly cycle, the student should be able to identify the main mathematical objects, state the essential definitions, solve representative problems, justify the principal steps, verify or interpret the result, and communicate the argument clearly.

Assessment in the exercise component focuses primarily on how the student works throughout the semester. Evidence of serious engagement includes:

- regular preparation and completion of the assigned problem sets,
- documented attempts to solve problems rather than bare answers,
- active participation in presentations and classroom discussion,
- separate, self-contained notes for individual problems,
- careful revision of work after feedback,
- increasing clarity, independence, and mathematical precision.

A student is not defined by the quality of the first solution or presentation. Some students may initially find mathematical notation, formal writing, or public explanation difficult. A weak beginning does not prevent a very good final grade for the exercise component. A student who works conscientiously, responds to feedback, corrects earlier weaknesses, and demonstrates substantial development may finish the exercise component with a very good grade.

This approach does not lower the standard of mathematical work. It recognises that reaching the standard through sustained effort and revision is itself a meaningful achievement. The exercise-component grade reflects a process of learning, not merely a snapshot taken at the beginning of the semester.

Systematic work according to these guidelines provides the basis for earning a passing grade for the exercise component. The examination is assessed separately and tests the student's mathematical knowledge and ability to use it independently. Regular work does not replace the examination, but it develops the understanding and skills that substantially increase the likelihood of successfully completing the course.

1.3 How to Prepare Mathematical Notes

A mathematical note is not merely a place where an answer is recorded. Its purpose is to show how a problem is understood, how relevant information is selected, how a method is chosen, and how a conclusion follows from definitions, assumptions, and earlier results.

Mathematical expressions are compact. A single formula may contain several ideas at once: objects, operations, conditions, and logical relationships. Reading a formula therefore means more than pronouncing its symbols. A student should be able to explain what the objects represent, what operation is being performed, why the expression is meaningful, and what information the formula provides.

The same principle applies to writing. A sequence of formulas without explanatory sentences rarely communicates a complete mathematical argument. Formulas should be connected by words that explain their role. A good solution tells the reader why a calculation is being performed, which definition or theorem is being used, whether its assumptions are satisfied, and how the result answers the original question.

1.3.1 Make the Note Self-Contained

Prepare one separate, self-contained note for each assigned problem, including all of its subparts. Begin by writing the problem or a complete and faithful formulation of it. The reader should not need to search through another document to discover what is being solved. After several weeks, you should still be able to open the note and understand both the question and the solution.

Identify what is given and what must be found, proved, or explained. Introduce the relevant notation and determine what kinds of mathematical objects appear in the problem. Before calculating, ask what information is available and what would count as a satisfactory answer.

1.3.2 Make the Reasoning Visible

Present the principal idea or method before, or while, carrying out the calculation. Important transitions should be explained with complete sentences. For example:

- Because the determinant is non-zero, the matrix is invertible.
- We may therefore multiply both sides by the inverse matrix.
- By the definition of the derivative, we consider the following limit.
- Since the direction vector is non-zero, it determines a line.
- Substituting the result into the original equation verifies the solution.

Such sentences are not decoration around the mathematics. They are part of the mathematics because they show the logical relationship between successive formulas.

It is not necessary to describe every elementary arithmetic operation. The amount of explanation should be proportional to the problem. A simple exercise may require only a few sentences, while a more substantial problem may require a longer argument. The goal is not to make every note long. The goal is to make every essential step understandable and justified.

Write as briefly as possible, but as completely as necessary.

1.3.3 A Complete Solution

A complete note should make the following five stages visible:

1. **Understand the problem.** Record the problem, identify the given information, and state clearly what must be found, proved, or explained.

2. **Choose a method.** Select the definitions, theorems, or techniques that connect the available information with the goal.

3. **Construct the argument.** Carry out the necessary steps and connect formulas with sentences that explain why each important transition is valid.

4. **Verify the result.** Check calculations, assumptions, domain restrictions, or geometric meaning whenever appropriate.

5. **State the conclusion.** Answer the original question clearly and in its proper context.

A result without its context is not a complete solution. A bare final answer, an unexplained sequence of formulas, or a few unconnected calculations will be treated as incomplete work and should be revised.

Quality cannot be replaced by quantity, but quality does not remove the obligation to complete the assigned work. Students are expected to prepare separate, complete solutions for the entire assigned problem set. Producing many unexplained answers is insufficient, but preparing one polished solution while ignoring the remaining problems is also insufficient.

1.3.4 Working with AI

We are studying in an age in which AI tools are widely available. The free versions of these tools are sufficient for the kinds of support expected in this course: asking questions, clarifying notation, generating simpler examples, checking reasoning, identifying missing steps, and improving the organisation of mathematical writing. Access to expensive software is not required.

AI should be used as an individual learning assistant. It may help transform rough calculations into clear mathematical prose, but it does not replace the student's mathematical work. Receiving a well-written answer is not the same as understanding it. The student remains responsible for checking every definition, calculation, assumption, conclusion, and formula included in the submitted note.

Use AI with a focused context. Work on one problem at a time and prepare one complete note for that problem. Do not paste an entire problem set into a single conversation and request all answers at once. Handling many unrelated tasks together encourages brief, generic, and mathematically weak notes. Focused work makes it easier to question each step, identify missing justifications, and produce a solution that can be understood and presented.

The one-problem-at-a-time principle does not mean that only one problem from the set should be prepared. Students are expected to repeat this focused process for every problem in the complete assigned set.

A polished AI-generated note is insufficient if the student cannot read its formulas, explain its terminology, justify its transitions, or present the argument independently. When using AI, ask not only for a solution, but also questions that deepen understanding:

- What does this formula mean?
- What mathematical object does each symbol represent?
- Why is this step valid?
- Which definition or theorem is being used?
- Are the assumptions of the theorem satisfied?
- Is there a simpler way to explain this transition?
- How can the result be checked?

- What should I be able to explain during class?

Preparing notes in this way is part of learning mathematics. It develops the ability to read notation, recognise mathematical objects, organise information, construct arguments, and communicate ideas precisely. Repeating this process for the complete problem set builds the habits of systematic and conscientious work that should accompany the student throughout university study and later professional life.

1.3.5 Student GitHub Workspace

The current instructions, problem templates, and technical workflow for student work are maintained in the public repository `dchorazkiewicz/Math_Problems_Repo`. Students should create a public fork of this repository and commit their notes to their own fork. The course repository is the common template; it is not the place where individual student solutions should be submitted.

The repository contains a short map of the working process:

- `README.md` explains the purpose and structure of the workspace and provides the weekly index;
- `SOURCE_MATERIAL.md` links the current lecture PDF and maps each week to the corresponding LaTeX problem source used by an AI assistant;
- `NOTE_GUIDELINES.md` defines the standard of a complete mathematical note;
- `AI_WORKFLOW.md` describes the one-problem-at-a-time workflow with AI, review, commits, presentation, feedback, and revision;
- `AGENTS.md` gives a connected AI assistant the operational rules for working safely in the student's fork;
- the `docs/` directory contains twelve weekly folders and one Markdown file for each problem.

The workspace uses Markdown for student notes, Git and GitHub for version history and public forks, MathJax for mathematical typesetting, MkDocs for the web presentation, and GitHub Actions for automatic validation and deployment. These instructions and templates may be updated independently of this PDF, so students should consult the repository for the current working procedure. The repository links back to this PDF and to its LaTeX sources. Students use the PDF as the readable course material containing the theory, definitions, examples, and problem sets. A connected AI assistant uses the current mapped LaTeX file as the exact machine-readable source of a selected problem statement.